1. Table of Contents

2. Claim Denial Classifier

2.1 Objective

The objective of this work is to rank insurance claims by their likelihood of denial, enabling a review team operating under fixed capacity to prioritise the claims most likely to be denied. The team can review the top 25% of ranked claims. **Performance is therefore measured using Recall@Top-25%**: of all claims that are ultimately denied, the proportion that appears within the top quartile of the ranked queue.

2.2 Results

The model achieves the following Recall@Top-25% across the three data splits:

Split	Recall@Top-25%	Lift vs Random
Training	54.91%	~2.2x
Validation	51.46%	~2.1x
Test	**50.71%**	**~2.1x**

Table 1: XGBClassifier evaluations on “train”, “validation”, and “test” splits.

Evaluated on unseen test data, the model achieves **50.71% recall at a 25% review rate**—a **2.1x improvement** over the random baseline (25% recall). Operationally, **this allows the review team to surface ~100% more denials** with no additional staffing or workload.

2.3 Predictions

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https://github.com/shivAM2886/ensemble-claim-denial/blob/main/data/current_claims_with_denial_probs.csv

2.4 Feature Selection

We initially evaluated feature plots to test how well individual variables delineate between denials and non-denials. While certain standalone features showed minimal predictive separation on their own, combining them unlocked significant signal. For instance, **has_prior_auth** and **prior_auth_required** showed only ~3% and ~7% differences in denial rates individually. However, combining them into an **unauthorized_procedure flag** (prior_auth_required == 1 and has_prior_auth == 0) widened the **denial rate separation to 27%.**

After testing both standalone variables and interaction terms, we validated feature impacts using SHAP values derived from an XGBoost classifier.

However, not all constructed features were retained. A subset, including higher timely-filing thresholds, visit-type and payer-type encodings, and several interaction and proxy features, was evaluated and subsequently excluded because each reduced validation recall. Features carrying limited signal introduce noise that degrades ranking stability. We therefore retained only those features that empirically improved recall.

The resulting feature set directly encodes the primary administrative conditions driving claim denials:

- **Unauthorized Procedure:** Prior authorization required but not obtained (prior_auth_required == 1 & has_prior_auth == 0).
- **Missing Documentation:** Required clinical or administrative documentation absent.
- **Missing Referral:** Referral required by payer rules but not present.
- **Out-of-Network Provider:** Provider rendering services outside the payer's network.
- **Unverified Eligibility:** Patient eligibility not confirmed prior to service delivery.
- **Patient Responsibility Delta:** Difference between total billed amount and expected payment, representing the uncovered portion of the claim.
- **Timely-Filing Flag:** Indicators for claims submitted beyond operational thresholds (e.g., >30 days).
- **Payer Identity:** Encoded indicators capturing payer-specific denial behavior and policy patterns.

Feature	Bucket	N (Claims)	Denials	Denial Rate
days_g30	1	343	99	28.86%
days_g30	0	1,779	349	19.62%

eligibility_not_verified_flag	1	269	95	35.32%
eligibility_not_verified_flag	0	1,853	353	19.05%

missing_documentation_flag	1	390	156	40.00%
missing_documentation_flag	0	1,732	292	16.86%

out_of_network_flag	1	389	100	25.71%
out_of_network_flag	0	1,733	348	20.08%

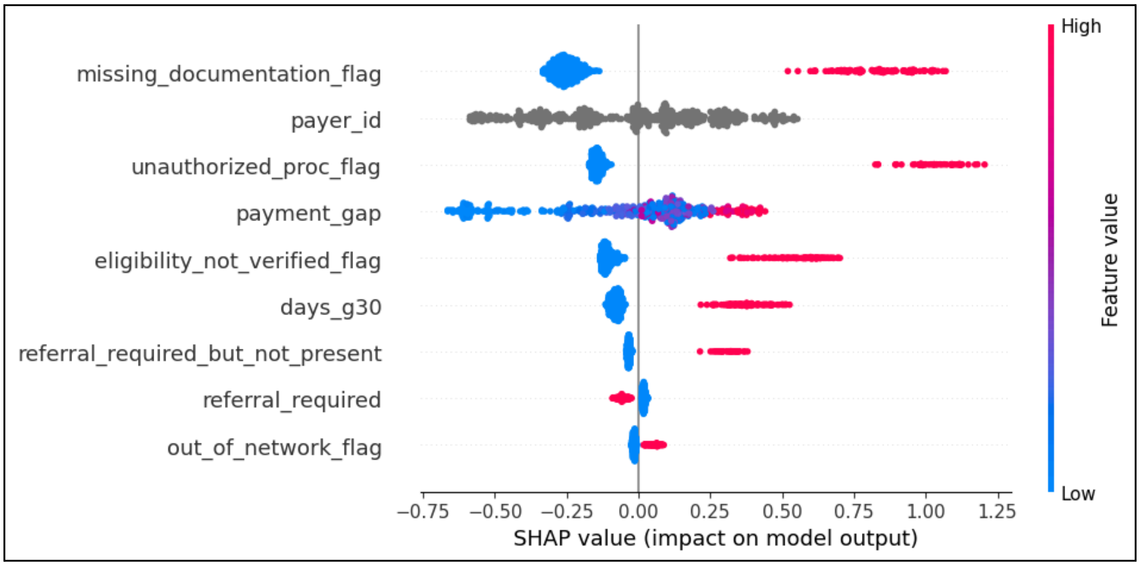
payer_id	P008	153	52	33.99%
payer_id	P012	169	49	28.99%
payer_id	P007	177	50	28.25%
payer_id	P006	187	47	25.13%
payer_id	P010	138	33	23.91%
payer_id	P005	198	44	22.22%
payer_id	P009	182	39	21.43%
payer_id	P003	171	34	19.88%
payer_id	P002	171	29	16.96%
payer_id	P004	135	20	14.81%
payer_id	P011	229	29	12.66%
payer_id	P001	212	22	10.38%

payment_gap	(8874.402, 67564.79]	425	119	28.00%
payment_gap	(5105.028, 8874.402]	424	99	23.35%
payment_gap	(3198.426, 5105.028]	424	90	21.23%
payment_gap	(1886.12, 3198.426]	424	86	20.28%
payment_gap	(211.809, 1886.12]	425	54	12.71%

referral_required_but_not_present	1	177	58	32.77%

referral_required_but_not_present	0	1,945	390	20.05%
unauthorized_proc_flag	1	205	92	44.88%
unauthorized_proc_flag	0	1,917	356	18.57%

Table 2: Denial Rates of individual & interaction features.



2.5 Modelling Approaches

The final model is an XGBoost gradient-boosted tree ensemble with hyperparameters tuned via Optuna to optimize Recall@Top-25% directly rather than relying on a proxy objective. To test whether performance could be pushed beyond this level, we evaluated a broad range of alternative approaches:

- **Additional Feature Engineering:** Adding coding-error proxies, payer interaction terms, and payment-to-procedure ratios produced no durable improvement.
- **Two-Model Architecture:** Scoring flagged claims (unauthorized_proc_flag == 1 & missing_documentation_flag == 1) separately from unflagged claims, then merging them into a single queue using various allocation rules, merely matched the single-model baseline.
- **Probability Ensembling:** Combining model outputs using maximum, mean, and logistic-regression stacking failed to improve ranking.
- **Alternative Objectives & Metrics:** Training with log-loss, PR-AUC, or a custom Recall@Top-25% objective and early-stopping metric yielded no lift.

- **Expanded Hyperparameter Search:** Pushing Optuna's search space beyond the original boundaries yielded no additional gains.

The fact that all evaluated architectures, features, and optimization strategies converged on this same performance ceiling provides strong evidence that the model has fully captured the available predictive signal.

2.6 Analysis of the Performance Ceiling

An error analysis of the denials ranked below the top-25% cutoff — those the model fails to surface — reveals two distinct populations.

- **Administratively defined denials**
    - Missing documentation, missing authorisation, eligibility failures, out-of-network providers, missing referrals, and late filing.

    - These conditions correspond directly to the engineered flags, and the model captures them reliably; they constitute the majority of claims correctly surfaced within the top 25%.

- **Medical necessity & coding denials**

Denial Reason	Train	Validation	Test
Payer policy or medical necessity issue	43.56%	48.00%	40.58%
Procedure or diagnosis coding error	10.40%	6.00%	14.49%
Total	53.96%	54.00%	55.07%

    - This group accounts for the majority of missed denials. Inspection confirms that these records carry no administrative flags, as they are driven by factors outside the current schema:
        - **Clinical Judgment:** Decisions regarding medical necessity based on diagnosis.
        - **Payer-Specific Coverage Rules:** Constraints such as formularies, frequency limits, and bundling policies.

    - Because models can only extract signals present in their input features, these non-administrative denials are inherently unpredictable under the current data structure. This explains why all modeling approaches converged: administrative denials are effectively captured, whereas medical-necessity and policy-driven denials cannot be recovered through feature engineering, ensembling, or hyperparameter tuning.

- **Conclusion:** The performance ceiling is dictated by the limitations of the available data, not the choice or execution of the model.

2.7 Path to Higher Performance

The performance constraint is **informational rather than algorithmic**, further gains depend on **enriching the data rather than refining the model**. Two additions would have the greatest impact:

- **Clinical context** — diagnosis-to-procedure appropriateness, prior treatments attempted, and severity indicators, which are the inputs used in medical-necessity determinations.

- **Payer coverage-policy data** — formularies, frequency limits, and bundling rules specific to each payer.

With these inputs, the **currently-unpredictable denials would become at least partially rankable**. In their absence, **the administrative signal available in the current dataset is substantially exhausted**.

2.8 Conclusion

- The model **achieves 51.70% Recall@Top-25% on held-out data, approximately 2 times the performance of random review for equivalent effort**.

- The remaining denials are **driven predominantly by clinical and payer-policy factors** that are **not represented in the available data**.

- The observed ceiling is a **data limitation rather than a modelling deficiency**, and the most effective route to improved performance is the incorporation of clinical and payer-policy information rather than additional model tuning.

3. Prompt

3.1 LLM Prompt (GPT-5.5)

```
You are a claims-denial risk assistant for healthcare billing analysts.
    Your job is to explain, in plain language, why a specific insurance claim is
    flagged as high risk for denial, based ONLY on the field values and risk
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drivers provided. Follow these rules strictly:

- Use ONLY the facts in the claim data below. Do NOT invent diagnoses,
amounts, dates, payer names, or any detail not present.
- Write for a busy analyst: plain language, no insurance jargon that would
need looking up (e.g., say "prior approval from the insurer" not "prior auth").
- Always include exactly ONE specific, actionable next step.
- Always state that this is a risk estimate, not a guarantee the claim will
be denied.
- Keep it to 2-3 sentences. No bullet points, no headers.


Model risk score (0-1, higher = more likely denied): 0.65

Claim facts (only use fields that are present/true):
- Prior approval required but not obtained: YES
- Required documentation missing: YES
- Required referral missing: no
- Provider is outside the insurer's network: no
- Patient eligibility not verified before service: no
- Uncovered amount (billed minus expected payment): $2348
- Filed later than the allowed window (e.g., >30 days): no

Write the 2-3 sentence explanation now.
```

3.2 LLM based Explanation for Top 10 High Risk Current Claims

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Find the explanations with features here:
https://github.com/shivAM2886/ensemble-claim-denial/blob/main/data/current_claims_top_10_risk_with_denial_probs_explanations.csv

claim_id	Predicted denial probability	explanation
CCLM-00372	0.65388024	This claim is high risk because prior approval from the insurer was required but not obtained, required documentation is missing, and the uncovered amount is $2,348. This is a risk estimate, not a guarantee the claim will be denied. Contact the insurer to ask whether they will accept an approval request after the service for this claim.

CCLM-00049	0.6169067	This claim is high risk because prior approval from the insurer was required but not obtained, required documentation is missing, and the required referral is missing; the uncovered amount is $3,480. This is a risk estimate, not a guarantee the claim will be denied. Next step: contact the insurer to ask whether the missing approval, documents, and referral can still be submitted for review.
CCLM-00087	0.61162937	The claim has a 0.61 denial risk estimate, not a guarantee that it will be denied. It is flagged because required documentation is missing, the provider is outside the insurer's network, patient eligibility was not verified before service, there is a $12,981 uncovered amount, and it was filed later than the allowed window. Upload the missing required documentation to the claim record before taking further action.
CCLM-00285	0.60293835	This claim is high risk with a score of 0.60 because prior approval from the insurer was required but not obtained, required documentation is missing, and the uncovered amount is $33,535. This is a risk estimate, not a guarantee the claim will be denied; next step: contact the insurer to ask whether retroactive prior approval can be accepted for this claim.
CCLM-00407	0.5945438	This claim is high risk because prior approval from the insurer was required but not obtained, required documentation is missing, and the uncovered amount is $16,890. This is a risk estimate, not a guarantee the claim will be denied; next step: submit the missing required documentation to the insurer.
CCLM-00177	0.58927286	This claim is flagged as higher risk because prior approval from the insurer was required but not obtained, required documentation is missing, and the claim was filed later than the allowed window. The $2,709 difference between the billed amount and expected payment also adds to the risk. This is a risk estimate, not a guarantee the claim will be denied; next step: confirm whether retroactive prior approval or an appeal package can be submitted with the missing documentation.
CCLM-00213	0.5848576	This claim is flagged as higher risk because prior approval from the insurer was required but not obtained, and the claim was filed later than the allowed window. The uncovered amount is $19,594, which may increase the financial impact if the insurer does not pay. This is a risk estimate, not a guarantee the claim will be denied; next step: check whether a late filing exception or retroactive prior approval request can be submitted.
CCLM-00023	0.5778236	This claim has a higher denial risk because prior approval from the insurer was required but not obtained, required documentation is missing, it was filed after the allowed window, and the uncovered amount is $3,269. This is a risk estimate, not a guarantee the claim will be denied; next step: prepare one appeal or corrected-claim packet that explains the late filing and includes any available supporting documentation for the missing approval and records.

CCLM-00344	0.57275003	This claim has a 0.57 denial risk estimate, mainly because prior approval from the insurer was required but not obtained, required documentation is missing, and $3,329 is not expected to be paid. This is only a risk estimate, not a guarantee the claim will be denied; next step: contact the insurer to ask whether they will accept a late prior approval request and the missing documentation for review.
CCLM-00307	0.5709316	This claim is flagged as higher risk because required documentation is missing, it was filed later than the allowed window, and the uncovered amount is $9,956. This is a risk estimate, not a guarantee the claim will be denied; next step: submit a late-filing appeal that includes the missing required documentation.